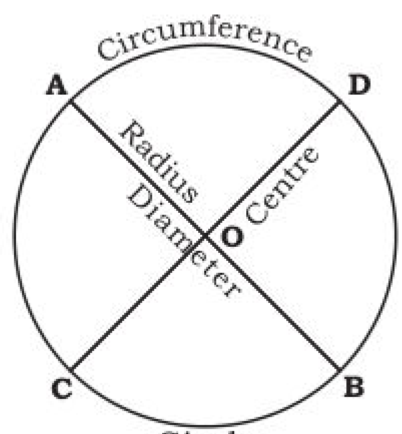
Unit 12

CIRCUMFERENCE, AREA AND VOLUME

12.1 CIRCUMFERENCE AND AREA OF CIRCLE

12.1 Circumference of a Circle

Circle is a familiar shape. O is centre of the circle. $\overline{OA}$ is radius. $\overline{AB}$ is diameter. **Circle** is a closed figure and every point on it is at the same distance from a given point O, the centre.



Circle

The measure of the length of the boundary of a circle is called its **circumference.**

Whereas AOD, BOD, AOC and BOC are sector of the circle. $\overline{OA}$, $\overline{OB}$, $\overline{OC}$ and $\overline{OD}$ all represent the radii of the given circle.

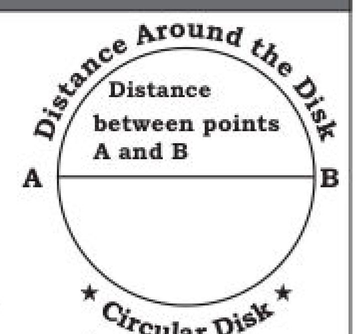
$\overline{AB}$ and $\overline{CD}$ are diameters. Since a diameter is a line segment, so it can easily be measured.

12.1.1 Express π as the ratio between the circumference and the diameter of a circle.

Activity: Let us work with a partner or in a group to find the ratio between the circumference and the diameter of a circle. We will need several circular objects, a calculator, a ruler, and a measuring tape and string.

Step 1

Choose one circular disk and answer the following. Consider a circular disk.



(a) Which do you think is greater:
the distance around the circular disk or
the distance between A and B

Let us try to find the answer of second question.

It is observed that the distance around the disk is greater.

(b) How many times as great do you think it is?

Step 2

Measure and record the distance around each disk (its **circumference**), to the nearest millimetre, using string.

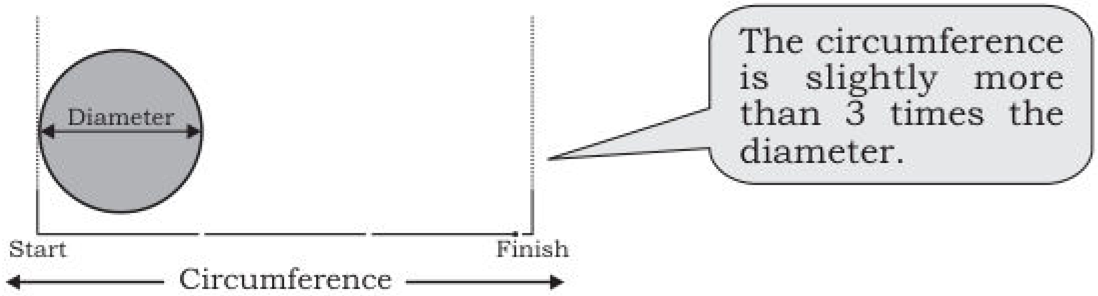
Step 3

Measure and record the distance across each disk (its **diametre**). Using ruler or tape.

1. Repeat Steps 2 and 3 for the other disk also. Record your results in a table given below: Include the results of your partner or other group members.

2. Also compare the numbers in the circumference column with those in the diameter column.

We find the ratio of circumference to diameter in all the above cases:



The table below shows the measure of diameters and approximate size of circumferences of four circles. Find the value of ratio $\left(\frac{\text{Circumference}}{\text{Diameter}}\right)$ for each circle. What did you notice?

Circle	Diameter	Circumference (Approximate value)	$\frac{\text{Circumference}}{\text{Diameter}}$ (Approximate value)
(i)	3.5 cm	11 cm	
(ii)	28 mm	88 mm	
(iii)	4.2 m	13.2 m	
(iv)	21 mm	66 mm	

Unit 12 CIRCUMFERENCE, AREA AND VOLUME

If we consider circles of different radii and find their circumferences, we will see that the ratio of each circumference to its diameter is approximately the same. This ratio is denoted by π.

The value of $\frac{\text{Circumference}}{\text{Diameter}}$ or π is the same for all circles. It is always about 3.14 or $\frac{22}{7}$.

Therefore $\boxed{\text{Circumference} = \pi \times \text{Diameter}}$

i.e. Circumference is π times greater than diameter.

12.1.2 Find the circumference of a circle using formula

There are different methods of measuring the circumference of a circle.

(a) Measurement of the circumference of a circle through activities.

Activity 1:

Steps: (i) Let circle is a cross section of a cylinder.

(ii) We want to measure the circumference of this circle.

(iii) Wrap a strip of paper or a string round the cylinder.

(iv) When the strip or the string reaches the initial end point, put a mark on that place.

(v) Measure the length of the strip or the string from the initial point to the place marked.

(vi) This is the measure of length of the circumference.

The perimeter of a circle is also known as the **circumference** of the circle.

Measurement of the circumference of a circle using formula.

We can use the following formula to find the circumference of a circle:

Circumference of the circle = $\pi \times$ Length of the diameter.

We know that diameter = $2\ r$ = Twice of radius.

Circumference $C = 2\pi r$, also $C = \pi d$, where d is the length of diameter.

Example 1. A circle has a radius of 35 mm. Find (a) its diameter, (b) its circumference. $\left(\text{Take } \pi = \frac{22}{7}\right)$

Solution:

(a) Diameter = 2 × Radius

$= 2 \times 35$ mm

$= 70$ mm

Its diameter is 70 mm.

(b) Circumference = π × Diameter

$= \frac{22}{7} \times \frac{70\text{mm}}{1}$

$= \frac{22 \times \cancel{70}^{10}}{\cancel{7}} = 220$ mm

Its circumference is 220 mm.

Example 2. The length of the diameter of a circle is 14 cm. Find the circumference of the circle?

Solution: The length of the diameterd = 14 cm and $\pi = \frac{22}{7}$

Thus, the circumference of the circle = π × d

$$= \frac{22}{7} \times 14 = \frac{22 \times \overset{2}{\cancel{14}}}{\underset{1}{\cancel{7}}} = 44 \text{ cm}$$

Example 3: Find the diameter and radius of the circle, when its circumference is 99 cm.

Solution: We know that:

Circumference = π × d

or $C = \pi d$

or $\frac{C}{\pi} = d$

So,

Diameter $= d = \frac{C}{\pi} = \frac{99}{\frac{22}{7}} = \frac{99 \times 7}{22}$

or $d = \frac{\overset{9}{\cancel{99}} \times 7}{\underset{2}{\cancel{22}}} = \frac{63}{2} = 31.5$ cm

and radius $= r = \frac{d}{2} = \frac{63/2}{2} = \frac{63}{4} = 15.75$ cm

EXERCISE 12.1

A. Find the circumference of the circle when its diameter is:

(1) d = 28 cm (2) d = 35 cm (3) d = 42 mm (4) d = 56 mm

B. Find the circumference of the circle when its radius is:

(1) 10.5 cm (2) 28 cm (3) 38.5 cm (4) 49 cm
(5) 59.5 cm (6) 63 cm (7) 80.5 cm (8) 77 cm

C. Find the radius of the circle when its circumference is:

(1) 22 cm (2) 66 cm (3) 88 cm (4) 110 cm

(5) 132 cm (6) 176 cm (7) 220 cm (8) 198 cm

D. Find the diameter of the circle, when its circumference is:

(1) 44 cm (2) 154 cm (3) 242 mm (4) 264 mm

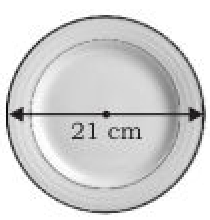
(5) 48.4 mm (6) 30.8 cm (7) 52.8 cm (8) 39.6 mm

E. Complete the table. Minute hand completes one round in 1 hour.

S.No.	Length of minute hand a clock is r	Find the distance covered by the minute hand
1.	r = 1.4 cm 	(i) Distance = $2\pi r$ $= \frac{2 \times 22 \times 1.4}{7} = 8.8$ cm
2.	r = 21 mm	
3.	r = 35 mm	
4.	r = 2.8 cm	
5.	r = 42 mm	
	(***Note:*** Minute hand completes one round in 1 hours)	

F. Find the minimum length of the string required to measure the circular boundary in each of the following:

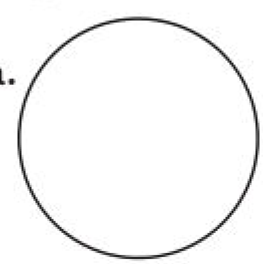
(1)



(2)

(3)

12.1.3 Find the area of a circular region using formula.

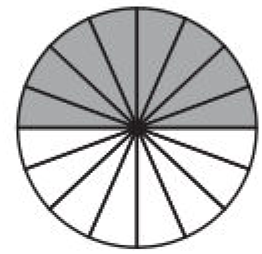


Let us find the Area of Circular Region through activity:

Activity:

(i) Draw a circle with suitable radius.

(ii) Cut the circle with a pair of scissors.



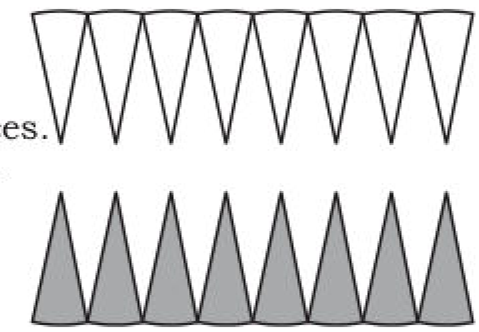
(iii) Fold the circle into two parts.

(iv) Keep on folding the circle, till we get 16 equal parts.

(v) Unfold the circle.

(vi) Shade half of the circle, as shown in the figure

(vii) Now cut the circle with a pair of scissors along the marks formed into 16 parts.

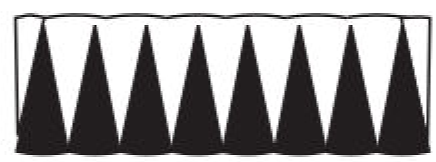


(viii) The length of every segment of the circle is equal to the radius.

(ix) Divide these 16 parts into two equal pieces.

(xi) Arrange the white parts cut in the upper half and the others in lower half.

(xii) The figure so formed is almost a rectangular region.



(xiii) The length of the rectangular region is equal to half of the circumference.

(xiv) The width of the rectangular region is equal to the radius of the circle.

(xv) The area of rectangular region is equal to the area of the circle.

Thus a circular region = Area of the rectangular region.

= Length of the rectangle × Breadth of the rectangle

= Length × Breadth

= (Half of the circumference) × (Radius of the circle)

$$= \left[\frac{1}{2}(2\pi r)\right] \times r$$

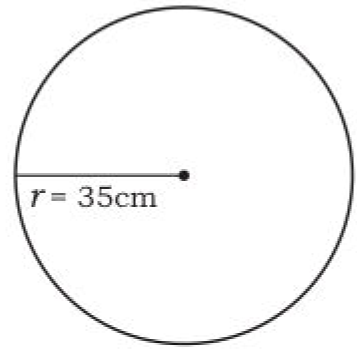
$$= \frac{1 \times \cancel{2} \times \pi r \times r}{\cancel{2}} = \pi r^2$$

Hence formula to find the area of a circular region is:

$$\boxed{\text{Area of a Circular Region} = \pi r^2}$$

where r is the radius of the circle.

Example 1: The radius of a circle is 35 cm. Find its area.



Solution: Area of the circular region = πr^2

Here radius = r = 35 cm; and $\pi = \frac{22}{7}$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times (35)^2 = \frac{22 \times 35 \times 35}{7}$$

$$= \frac{22 \times 35 \times \cancel{35}^{5}}{\cancel{7}_{1}} = 3,850 \text{ sq. cm}$$

Example 2: The area of a circular region is 1,386 sq. cm. Find its radius.

Solution: Given area A of the circular region is 1386 sq. cm.
We know that the formula for finding area of circular region is: $A = \pi r^2$.

$$\text{Therefore } 1386 = \pi r^2$$

$$\text{or } \frac{22}{7} r^2 = 1386$$

$$\text{or } r^2 = 1386 \div \frac{22}{7}$$

$$\text{So, } r^2 = \frac{1386 \times 7}{22} = \frac{11 \times 63 \times 2 \times 7}{22}$$

$$\text{Therefore } r^2 = 63 \times 7 = 3 \times 3 \times 7 \times 7$$

$$\text{Thus radius} = r = 3 \times 7 = 21 \text{ cm}$$

EXERCISE 12.2

A. Find the area of circular region when its radius is:

(1) 14 cm (2) 10.5 cm (3) 21 cm (4) 17.5 mm

(5) 35 mm (6) 24.5 cm (7) 71.4 cm (8) 38.5 cm

B. Find the radius of the circular region when its area is:

(1) 154 sq. mm (2) 2,464 sq. cm (3) 7546 sq. mm

(4) 5,544 sq. mm (5) 6.16 sq. mm (6) 962.5 sq. cm

(7) 38.5 sq. m (8) 75.46 sq. mm (9) 13.86 sq. m

C. Find the area of circular region when its diameter is:

(1) 21 cm (2) 28 cm (3) 42 cm (4) 56 cm

(5) 8.4 cm (6) 9.8 cm (7) 11.2 cm (8) 12.6 cm

D. Find the diameter of the circular region when its area is as under:

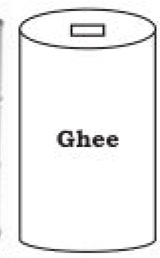
(1) 154 cm^2 (2) 1,386 (mm^2) (3) 3,850 (mm^2) (4) 9,856 mm^2

(5) 124.74 cm^2 (6) 186.34 cm^2 (7) 221.76 cm^2 (8) 260.26 cm^2

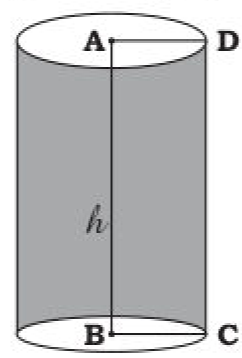
12.2 SURFACE AREA AND VOLUME OF A CYLINDER

Cylinders: We have already learnt about cylinders in previous class. Ghee tins, cooking oil tins, cold drink tins, drums of tarcol etc are the examples of cylinder.


Sound Drum



Activity: Staking of the coins. Let us keep on putting one rupee, two rupee and five rupee coins one above the other. What type of shapes, we get?



12.2.1 Find surface area of a cylinder using formula.

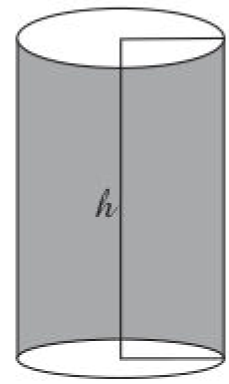
Surface of cylinder consists of three parts.
Two equal circular surfaces and a third curved surface.

Let us find the area of these three surfaces.
Let the radius of each of the circular surface be r and height of cylinder surface be h.

(i) Area of the two circular regions of the cylinder = Area of the upper base of the cylinder + Area of the lower base of the cylinder

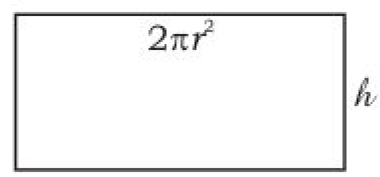
$= \pi r^2 + \pi r^2 = 2\pi r^2$

(ii) To find the area of the curved surface of the cylinder: We Take sheet of paper whose width is equal to the height of the cylinder.



Wrap it round the cylinder and cut of the extra piece, this will take the shape of the rectangle. The width of this paper will be h and the length will be equal to the circumference of the base, the circular region of the cylinder.

Area of the curved surface of the cylinder



$= $ Area of the rectangle

$= $ Length $\times$ Breadth

$= 2\pi r \times h$

$= 2\pi rh$

Thus the total area of the surface of the cylinder

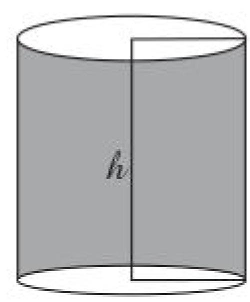
$= $ The area of the two circular surfaces

$+$ Area of the curved surface of the cylinder.

$= 2\pi r^2 + 2\pi rh$

$= 2\pi r(r + h)$

Thus the surface area of the cylinder $= 2\pi r(r + h)$



Example 1: The radius of the base of a cylinder is 14 cm. And the height of the cylinder is 20 cm. Find the total surface area of the cylinder.

Solution: Radius of the base $= r = 14$ cm

Height of cylinder $= h = 20$ cm $\left(\text{Take } \pi = \frac{22}{7}\right)$

Formula for finding the total surface area of the cylinder $= 2\pi r(r + h)$

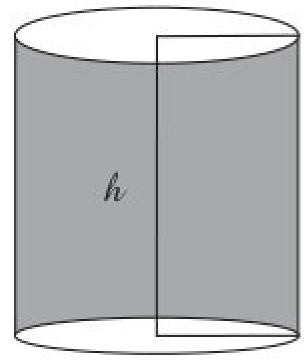
$$\text{i.e. } A = 2\pi\, r(r + h) = 2 \times \frac{22}{7} \times 14\,(14 + 21)$$

$$= \frac{2 \times 22 \times \overset{2}{\cancel{14}}}{\underset{1}{\cancel{7}}} \times (35)$$

$$= 88 \times 35 = 3{,}080 \text{ sq. cm}$$

Hence the total surface area of the sylinder is 3,080 cm^2.

Example 2: Find the height of a cylinder when its radius is 10.5 cm. Total surface area is 1650 cm^2.



Solution: Height of the cylinder $= h = ?$

radius of circular base $= r = 10.5$ cm

A = Total surface area = 1650 cm^2

Formula: $A = 2\pi r\,(r + h)$

$$\text{or } 1650 = 2 \times \frac{22}{7} \times 10.5\,(10.5 + h)$$

$$\text{or } 1650 = \frac{2 \times 22 \times \overset{1.5}{\cancel{10.5}}}{\underset{1}{\cancel{7}}} \times (10.5 + h)$$

$$\text{or } 1650 = 693 + 66h$$

$$\text{or } 1650 - 693 = 66h$$

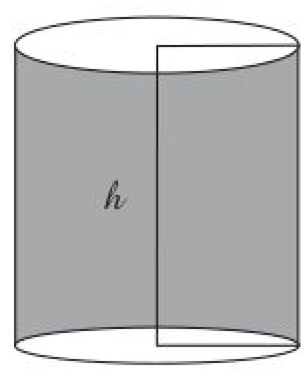
$$\text{or } 957 = 66h$$

$$\text{or } h = \frac{957}{66} = \frac{\overset{\overset{29}{\cancel{319}}}{\cancel{957}}}{\underset{\underset{2}{\cancel{22}}}{\cancel{66}}} = \frac{29}{2} = 14.5$$

$$\text{or } h = 14.5 \text{ cm}$$

Thus the required height of the cylinder is 14.5 cm

Example 3: Find the value of radius and height of a cylinder whose total surface area is 2200 sq. cm. And the sum of radius and height is 35 cm.



Solution: Total surface area = A = 2,200 cm^2

A = Total surface area = 2,200 cm^2

$r + h = 35$ cm

We have to find the value of radius and height of cylinder.

Formula: $A = 2\pi r\,(r + h)$

$$2200 = 2 \times \frac{22}{7} \times r\,(35)$$

or $2200 = \left(\frac{2 \times 22 \times 35}{7}\right) r$

or $2200 = \frac{44 \times \overset{5}{\cancel{35}}}{\underset{1}{\cancel{7}}} r$

or $2200 = 220r$

or $\frac{2200}{220} = r$

or $10 = r$ or $r = 10$ cm

Therefore radius $= r = 10$ cm

Again $r + h = 35$

$10 + h = 35$

Therefore $h = 35 - 10 = 25$ cm

Thus the required radius of the cylinder is 10 cm and height is 25 cm.

EXERCISE 12.3

A. Find the total surface area of the cylindrical region when the radius of the base of cylinder and height of the cylinder are given.

(1) Radius = 14 cm, height = 26 cm
(2) Radius = 10 cm, height = 18 cm
(3) Radius = 21 mm, height = 29 mm
(4) Radius = 17.5 cm, height = 22.5 cm
(5) Radius = 30 cm, height = 40 cm
(6) Radius = 25 mm, height = 41.5 mm

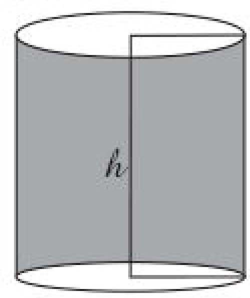
B. Find the height of the cylindrical region when its radius and total surface area are given.

(1) Radius = 10.5 cm, total surface area = 1980 cm^2
(2) Radius = 17.5 cm, total surface area = 4400 cm^2
(3) Radius = 12 mm, total surface area = 2112 mm^2
(4) Radius = 15 cm, total surface area = 3960 cm^2
(5) Radius = 28 cm, total surface area = 10560 cm^2

C. Solve the following.

1. Find the value of radius and height of a cylinder. The sum of its radius and height is 28 cm. Also total surface area of the cylinder is 1408 cm^2.
2. The sum of radius and height of a cylinder is 35 cm. The total area of the cylinder is 3,300 cm^2. Find its radius and height separately.
3. Find the value of radius and height of a cylinder when sum of its radius and height is 49 cm and total surface area is 5,852 cm^2.
4. The sum of radius and height of a cylinder is 56 mm. The total surface area is 6,864 $(mm)^2$. Find the radius and height of the cylinder.

II. Find the volume of a cylindrical region using formula



We know that the base of a cylinder is a circular surface. We denote its radius by r. We have already learnt to find the area of the circular region.

Area of circular region = Area of the base of the cylinder

$$\mathbf{A} = \pi r^2$$

If the height of the cylinder be denoted by h, then:

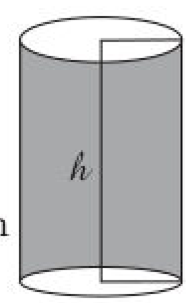
Volume of cylinder = Area of circular region × height of cylinder

$$\mathbf{V} = \pi r^2 \times h = \pi r^2 h$$

Hence the volume of the cylinder = $\pi r^2 h$ where r represents the radius of the circular base and h the height of the cylinder.

Example 1: If the length of the diameter of a cylinder is 84 cm and height is 55 cm; what is its volume?

Solution:



d = Diameter of the given cylinder = 84 cm

Therefore the radius r of the given cylinder = $\frac{d}{2} = \frac{84}{2}$ cm

Thus r = 42 cm

Now formula for finding volume of the cylinder is V = $\pi r^2 h$

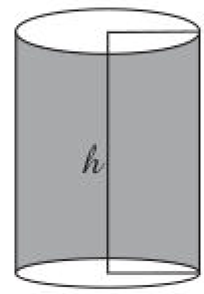
Then $V = \frac{22}{7} \times (42)^2 \times 55$

$= \frac{22}{\cancel{7}_1} \times \cancel{42}^{6} \times 42 \times 55$

$= 22 \times 6 \times 42 \times 55 = 304,920$ cubic centimetres

Example 2: The volume of a cylinder is 184,800 cubic centimetres and the radius of the circular base of the cylinder is 35 cm. Find the height

Solution: The formula of volume of the cylinder is $V = \pi r^2 h$



$V = 184,800$ cubic centimetres

$r = 35$ cm

$h = ?$

$184800 = \frac{22}{7} \times (35)^2 \times h$

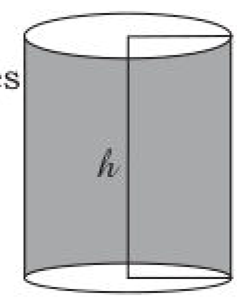
$\frac{7 \times 184800}{22 \times 35 \times 35} = h$

$h = \frac{\cancel{7} \times \cancel{184800}^{\cancel{7392}\ \cancel{1056}\ \cancel{528}\ 48}}{\cancel{22}_{\cancel{11}\ 1} \times \cancel{35}_{\cancel{7}\ 1} \times \cancel{35}_{\cancel{7}\ 1}} = 48$ cm

Thus the required height of the given cylinder is 48 cm.

Example 3: Find the radius of the cylinder, when its volume is 4,400

Solution:



V = Volume of the cylinder = 4,400 cubic centimetres

h = Height of the cylinder = 14 cm

r = radius of circular base of the cylinder = ?

Formula for finding volume of cylinder = $V = \pi r^2 h$

$4,400 = \frac{22}{7} \times r^2 \times 14$

$4,400 = \frac{22 \times \cancel{14}^{2}}{\cancel{7}_1} r^2$

$4400 = 44r^2$

or $r^2 = \frac{4400}{44} = 100$

Therefore $r = \sqrt{100} = 10$

Hence radius of circular base of the cylinder is 10 cm.

EXERCISE 12.4

A. In each of the following, the radius of the base of cylinder and its height is given. What is the volume in each case?

(1) Radius = 14 cm and height = 20 cm
(2) Radius = 25 cm and height = 35 cm
(3) Radius = 28 cm and height = 40 cm
(4) Radius = 20 cm and height = 28 cm
(5) Radius = 1.05 m and height = 2.5 m
(6) Radius = 63.5 cm and height = 75 cm
(7) Diameter = 70 mm and height = 50 mm
(8) Diameter = 60 mm and height = 63 mm

B. In each of the following, the radius of the base of the cylinder and its volume is given. What is the height of cylinder in each case?

(1) Radius = 14 cm and volume = 15,400 cubic centimetres
(2) Radius = 20 cm and volume = 35,200 cubic centimetres
(3) Radius = 25 mm and volume = 82,500 cubic millimetres
(4) Radius = 21.5 mm and volume = 305,085 cubic millimetres
(5) Radius = 30 mm and volume = 99,000 cubic millimetres
(6) Radius = 28 cm and volume = 98,560 cubic centimetres

C. In each of the following, the height of the cylinder and its volume is given. What is the radius of the base of the cylinder in each case?

(1) Height = 25 cm and volume = 34,650 cubic centimetres
(2) Height = 20 m and volume = 12,320 cubic metres
(3) Height = 28 cm and volume = 35,200 cubic centimetres
(4) Height = 42 cm and volume = 118,800 cubic centimetres
(5) Height = 56 mm and volume = 281,600 cubic millimetres
(6) Height = 30 cm and volume = 41,580 cubic centimetres

12.2.3 (a) Solve real life problems involving Circumference and Area of Circle

Example 1: The radius of the wheel of a machine is 84 cm. How much wire will be needed to wound round ten full rounds.

Solution:

r = radius of the wheel = 84 cm = 0.84 m

Formula for finding circumference is $C = 2\pi r$

Thus we have circumference = 5.28 m

The wire required for one round is 5.28 m

Therefore the wire required for ten rounds is (5.28×10)m = 528 m

$$C = \left(2 \times \frac{22}{7}\, 0.84\right) \text{m}$$

$$\text{or } C = \frac{2 \times 22 \times 0.84 \text{ m}}{7}$$

$$\text{or } C = \frac{2 \times 22 \times \overset{0.12}{\cancel{0.84}} \text{ m}}{\underset{1}{\cancel{7}}}$$

$$\text{or } C = 5.28 \text{ m}$$

Hence the length of the required wire is 528m.

Example 2: The diameter of a round table is 4.2 m. How many metres of lace are required for its border? Also find its cost at the rate Rs 40 per metre.

Solution:

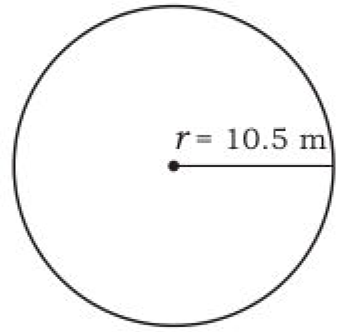
First we have to find the circumference of round table.

Circumference = $\pi\, d = \frac{22}{7} \times 4.2 \text{ m} = \frac{22}{\cancel{7}} \times \overset{0.6}{\cancel{4.2}} = 22 \times 06 = 13.2$ m

Thus the length of the lace required for round the circular cloth is 13.2 metres.

The cost of the length of lace = Rate × length
= Rs 40 × 13.2
= Rs 528

Example 3: What is area of a circular floor and the cost of flooring it at the rate of Rs 150 per square metre. When the radius of circular floor is 10.5 m.



Solution: The radius of circular floor is 10.5 m.

Formula for finding area of circular floor is πr^2

i.e. $A = \pi r^2 = \frac{22}{7} \times 10.5 \times 10.5$

$$\text{or } A = \frac{22 \times \overset{1.5}{\cancel{10.5}} \times 10.5}{\underset{1}{\cancel{7}}} = 346.5 \text{ m}^2$$

Thus the area of circular floor is 346.5 m^2.

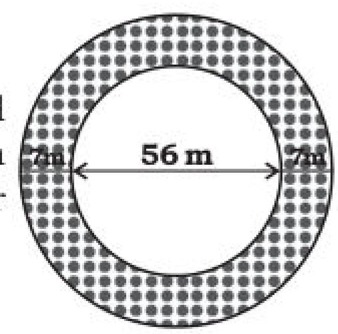
The cost of flooring it is: Rate × Area

= Rs (150 × 346.5)

= Rs 51975 . 00

Hence cost of flooring the circular floor is Rs 51,975.

Example 4: The diameters of two circular grounds and the width of the road encircling it is given in the figure. Find the area of the road and cost for cementing it a Rs 50 m^2.



Solution: There are two circles:

(i) Outer circle having diameter = (56 + 14) m = 70 m

Therefore the radius of outer circle = $\frac{70}{2}$ m = 35 m = R

(ii) Inner circle having diameter = 56 m

Therefore the radius of inner circle = 28 m = r

Now we find the area of both circles.

Area of outer circle = $\pi R^2 = \frac{22}{7} \times 35 \times 35 = \frac{22 \times \overset{5}{\cancel{35}} \times 35}{\underset{1}{\cancel{7}}} = 3{,}850 \text{ m}^2$

Area of inner circle = $\pi r^2 = \frac{22}{7} \times 28 \times 28 = \frac{22 \times \overset{4}{\cancel{28}} \times 28}{\underset{1}{\cancel{7}}} = 2{,}464 \text{ m}^2$

Hence area of road = Area of outer circle – Area of inner circle

$$= 3{,}850\ m^2 - 2{,}464\ m^2 = 1{,}386\ m^2$$

The cost of cementing the road is: Rate × Area of the road

$$= Rs\ 50 \times 1386 = 69{,}300 \text{ rupees}$$

Hence the cost of cementing the circular road is Rs 69,300.

EXERCISE 12.5

1. The diameter of a circular garden is 70 m. The garden has 5 rounds of wire fencing. If the wire costs Rs 10 per metre. Find

2. Find the radius of a circular ring prepared from a wire used for a square with side 22 cm.

3. The radius of a bicycle wheel is 42 cm. Each wheel completed 1000 rotations when Ahmed travelled from home to school. Find

4. The total cost of the wire required for four rounds of a circular fencing is Rs 3,960/- If the wire costs Rs 5 per metre; find the

5. A piece of wire is bent in the shape of an equilateral triangle, having each side 13.2 m. It is again re-bent to form a circular ring.

6. What is the area of a circular garden and the cost of gardening it at the rate of Rs 20 per sq. metre? When the radius of the

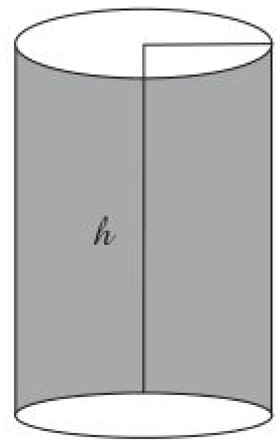
7. There are two circular grounds. The outer circular ground has diameter 49 m. The inner circular ground has diameter 35 m. Find the area of the road and the cost of cementing it at Rs 100

8. There is a round wooden table of 2.8 m diameter. Find the area of the surface of the table and the cost of polishing it at the rate

9. A circular fountain is built in the centre of crossing. Its radius is 3.5 m. What is the area of the fountain? Also find the cost of

10. A table cover is made to cover a round dinning table. The diameter of the table cover is 3.5 m. What is area of the table cover? Also

12.2.3 (b) Solve real life problems involving Surface Area and Volume of a Cylinder



Example 1: An open cylindrical tank has a circular base of diameter 2.1 m. The height of the tank is 4 m. Find how many square metres of steel sheet are required for the tank. Also find the cost of tank at the rate of Rs 400 per m^2.

Solution:

Diameter of circular base = 2.1 m

Radius of circular base = $r = \frac{2.1}{2}$ m

or $r = 1.05$ m

Area of circular base = Area of circle = πr^2

$$A = \pi r^2$$

$$= \left(\frac{22}{7} \times 1.05 \times 1.05\right) m^2 = \left(\frac{22 \times \overset{0.15}{\cancel{0.15}} \times 1.05}{\cancel{7}_1}\right) m^2$$

$$= (22 \times 0.15 \times 1.05)\ m^2 = (3.30 \times 1.05)\ m^2 = 3.465\ m^2$$

Curved surface area of cylinder $= 2\pi rh = \left(2 \times \frac{22}{7}\ 1.05 \times 4\right) m^2$

$$= \left(\frac{2 \times 22 \times \overset{0.15}{\cancel{1.05}} \times 4}{\cancel{7}_1}\right) m^2 = (44 \times 0.60)\ m^2 = 26.40\ m^2$$

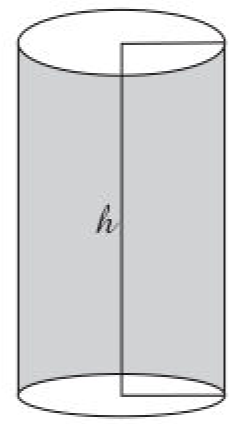
Now total area of steel sheet required to make the tank is

$$= 3.465\ m^2 + 26.40\ m^2$$

$$= 29.865\ m^2$$

The cost of tank = Rate × Area = 400 × 29.865 = Rs 11946.000 = Rs 11,946.

Example 2: The radius of a circular well is 1.4 m and its depth is 20 m. What will be the cost of designing the well at the rate of Rs 300 per cubic metre?



Solution:

Radius of the base of the circular well = r = 1.4 m

Depth of the well = h = 20 m

Volume of the well = V = $\pi r^2 h$

$$\text{Volume} = V = \frac{22}{7} \times 1.4 \times 1.4 \times 20 = \frac{22 \times \overset{0.2}{\cancel{1.4}} \times 1.4 \times 20}{\underset{1}{\cancel{7}}} = 4.4 \times 28.0$$

$$= 123.2 \text{ cubic metres}$$

Cost of designing = Rate × Volume

= 300 × 123.2 = Rs 36960.0 = Rs 36,960

Hence the volume of the well = 123.2 cubic metres and the cost of

EXERCISE 12.6

1. The radius of an open pond in a cylindrical shape is 10.5 m and depth is 8 m. Find the cost of cementing its floor and curved

2. Find the total surface area of a cylidrical steel tank 8.4 m long. The radius of its circular base is 7 m. Also find its cost at the rate of Rs 100 per sq. metre.

3. An open cylindrical tank has a base of diametre 2.8 m. The height of the tank is 5m. Find how many square metres of steel sheet are required for the tank. Also find the cost of the tank at

4. The length of a steel pipe is 2.8 m and radius 7 cm. Calculate its

5. Find the total surface area of oil drum whose length is 1.5 m and

6. Find the capacity of a circular water tank in litres when the height of the tank is 5 m and its diameter is 4.2 m. (Hint: Convert metres into centimetres and 1 litre = 1000 cm^3).

7. A cylindrical tin can is 63 cm high and its radius is 25 cm. Find

8. A well is 16 m deep and its diameter is 7 m. How much soil is

9. A tin pack of a soft drink is 10.5 cm long and the radius of the

10. The diameter of a circular well is 3.5 m and its depth is 24 m. What will be the cost of digging it at the rate of Rs 250 per

11. Find the radius of an oil drum whose volume is 11 m^3.

12. The capacity of an oil drum is 3850 litres. What is its volume in cm^3. Find its height in metres when its radius is 70 cm.

13. The internal diameter of a round cylindrical shape mosque is 28 m and height of curved walls is 8.75 m. Find the cost of tiles using on the walls only at the rate of Rs 200 per m^2.

14. The internal diameter of a swimming pool is 42 m. The depth is 5 m. Calculate the cost of tiles using round the wall at the rate of Rs 100 per m^2.

REVIEW EXERCISE 12

A. Answer the following:

1. What is circumference of a circle. Give formula to find circumference.
2. What is relation between diameter and circumference?
3. What is diameter of a circle? Give relation between radius and diameter.
4. What is radius of a circle? Give formula for finding area of a circle.

5. Write the formula for total surface area of a cylinder.
6. What is volume of a cylinder? Express its formula.
7. What is the formula for finding diameter from circumference of a circle?
8. In a cylinder when radius = height, what is the formula for finding volume of a cylinder?

B. Fill in the blanks:

1. The formula for finding curved surface area of the cylinder is ____________.
2. Tin pack of soft drink is an example of ___________.
3. Take ten coins of Rs 5 and stake up a file then ___________ type of shape is obtained.
4. When radius of cylinder is in metres; the unit for volume of cylinder will be _____________.
5. The value of π upto three decimal places is ___________.

C. Solve the following mentally and (✓) the correct answer.

1. The circumference of a circle is ____ cm when its diameter is 7 cm.

   (i) $\frac{22}{7}$ (ii) $\frac{7}{22}$ (iii) 7 (iv) 22

2. The area of a circle is ________ sq. cm when its radius is 1 cm.

   (i) $\frac{22}{7}$ (ii) $\frac{7}{22}$ (iii) $\frac{1}{7}$ (iv) $\frac{1}{22}$

3. The area of circular base of cylinder is ___ sq.cm when the value of $r^2 = \frac{1}{22}$ sq.cm.

   (i) 7 (ii) $\frac{1}{7}$ (iii) 22 (iv) $\frac{1}{22}$

4. Area of curved surface of a cylinder is _______cm^2 when $r = 7$ cm and $h = \frac{1}{22}$cm.

   (i) 1 (ii) 2 (iii) 3 (iv) 4

5. Total surface area of a cylinder is __________ cm^2 when $r = 1$ cm and $h = 1$ cm.

   (i) 2π (ii) 3π (iii) 4π (iv) π

6. Volume of a cylinder is ______ cm^3 when $r = 1$ cm and $h = 1$ cm.

   (i) $\frac{22}{7}$ (ii) $\frac{7}{22}$ (iii) 22 (iv) 7

7. Area of circular upper base and lower base of cylinder is ___ cm^2 when $r = 1$ cm.

(i) $\frac{22}{7}$ (ii) $\frac{44}{7}$ (iii) $\frac{7}{22}$ (iv) $\frac{7}{44}$

8. Height of an oil drum is 1 m and radius of the drum is also 1 m. Then its volume is _______ m^3.

(i) 7 (ii) 22 (iii) $\frac{22}{7}$ (iv) $\frac{7}{22}$

9. The length of the minute hand of a clock is 3.5 cm. The distance covered in 1 hour is ________cm.

(i) 22 cm (ii) 7 cm (iii) 3.5 cm (iv) 2.2 cm

10. In a circle the radius is increased by twice in length. The area of the circle will be increased by ____________.

(i) Twice (ii) Thrice (iii) Four times (iv) None of these

SUMMARY

- Circumference is the length of the boundary of a circle.
  Circumference $C = 2\pi r$ or $C = \pi d$
- Ratio between circumference and diameter of a circle is denoted by π i.e $\frac{\text{Circumference of the circle}}{\text{Diameter of the circle}} = \frac{22}{7} = 3.14 = \pi$.
- Area of a circle is the area of the circular region that is the number of square units inside the circle. $A = \pi r^2$ square units.
- The surface area of a cylinder consists of three parts; two equal circular surfaces and a third curved surface. Total surface area of cylinder = $\pi r^2 + \pi r^2 + 2\pi rh = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$
- Volume of a cylinder = Area of the circular region × Height of the cylinder.

$$V = \pi r^2 \times h$$
$$V = \pi r^2 h \text{ cubic units}$$

Unit 13 INFORMATION HANDLING

Introduction

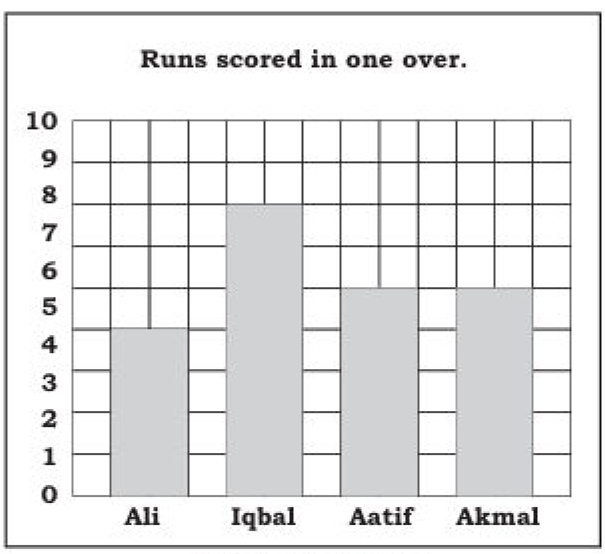
Information handling plays an important role in the subject of statistics. This subject has been constantly expanding during the last few centuries and its definition is also changing. This subject originated due to the desires of the rulers in ancient times to find how much manpower was available in their kingdoms. For this they used to hold census in an organized manner. This way they used to be well informed about their power and the capability to wage wars. That is why statistics used to be called the rulers' science and political arithmetic.

13.1 FREQUENCY DISTRIBUTION

13.1.1 Demonstrate data presentation

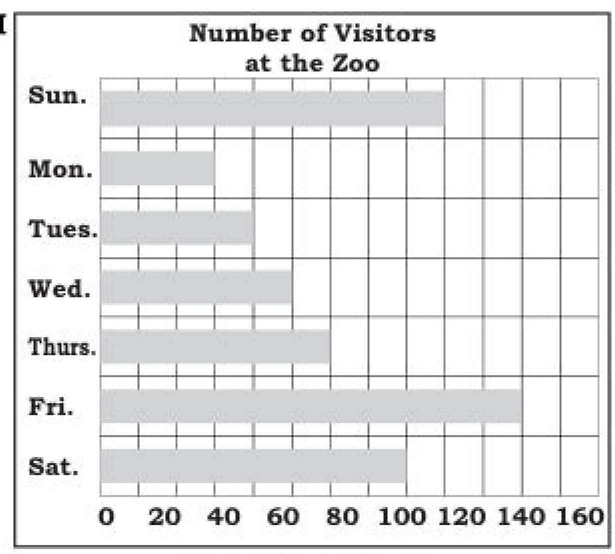
In our daily life we collect information and present it in different ways bar graphs, charts etc so that it could be understood easily and the useful conclusions can be drawn.

I



Names of Players

II



Number of Visitors

In previous class, we have learnt that 'Data' means, the facts about information that are the results of measurement, observation or experiments.